



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**PHYSICAL SCIENCES: CHEMISTRY (P2)
FISIESE WETENSKAPPE: CHEMIE (V2)**

2021

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 19 pages./
*Hierdie nasienriglyne bestaan uit 19 bladsye.***

QUESTION 1/VRAAG 1

- | | | |
|------|------|-------------|
| 1.1 | C ✓✓ | (2) |
| 1.2 | D ✓✓ | (2) |
| 1.3 | C ✓✓ | (2) |
| 1.4 | B ✓✓ | (2) |
| 1.5 | D ✓✓ | (2) |
| 1.6 | C ✓✓ | (2) |
| 1.7 | B ✓✓ | (2) |
| 1.8 | B ✓✓ | (2) |
| 1.9 | A ✓✓ | (2) |
| 1.10 | B ✓✓ | (2) |
| | | [20] |

QUESTION 2/VRAAG 2

- 2.1
- 2.1.1 F ✓ (1)
- 2.1.2 B & F ✓ (1)
- 2.1.3 C ✓ (1)
- 2.2
- 2.2.1 Haloalkane / alkyl halide ✓
Haloalkaan/alkielhalied (1)
- 2.2.2 3,5-dibromooctane ✓✓✓
3,5-dibroomoktaan
- Marking criteria/Nasienkriteria:**

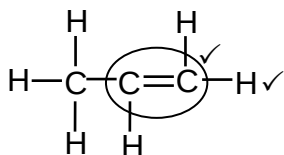
 - Octane/Oktaan ✓
 - Dibromo/Dibroom ✓
 - Substituents (dibromo) correctly numbered, hyphens, commas correctly used./
Substituente (dibroom) korrek genommer, koppeltekens en kommas korrek gebruik. ✓
- (3)
- 2.3
- 2.3.1 Pentan-3-one ✓✓
Pentan-3-oon
OR/OF
3-pentanone ✓✓
3-pentanoon
- Marking criteria/Nasienkriteria:**

 - Pentanone/pentanoon ✓
 - Correct position of functional group. ✓
Korrekte posisie van funksionele groep.
- (2)
- 2.3.2 3-methyl✓butan-2-one✓/3-metielbutan-2-oon
OR/OF
3-methyl✓butanone✓/3-metielbutanoon
OR/OF
methyl✓butanone✓/metielbutanoon
OR/OF
3-methyl✓-2-butanone✓/3-metiel-2-butanoon (2)
- 2.4
- 2.4.1 Hexyl✓methanoate ✓
Heksielmetanoaat (2)
- 2.4.2
- $$\begin{array}{c} \text{O} \\ || \\ \text{---C---O---C---} \\ | \end{array}$$

OR/OF

$$\begin{array}{c} \text{O} \\ || \\ \text{R---C---O---R} \end{array}$$
- (1)
- 2.5
- 2.5.1 Cracking/Elimination ✓
Kraking/eliminasië (1)
- 2.5.2 C₇H₁₆ ✓✓ (2)

2.5.3



Notes/Aantekeninge

- Functional group/Funksionele groep: ✓
- Whole structure correct/Hele struktuur korrek: ✓

(2)
[19]

QUESTION 3/VRAAG 3

3.1

Marking guidelines/Nasienkriteria:

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark./Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The pressure exerted by a vapour at equilibrium with its liquid in a closed system. ✓✓

Die druk uitgeoefen deur 'n damp in ewewig met sy vloeistoffase in 'n geslote sisteem.

(2)

3.2

Functional group/Type of intermolecular forces/Homologous series ✓

Funksionele groep/Tipe intermolekulêre kragte/Homoloë reeks

(1)

3.3

B ✓

(1)

3.4

Marking criteria/Nasienkriteria

- State hydrogen bonding in **A**./Noem waterstofbinding in **A**. ✓
- State dipole-dipole forces in **B**./Noem dipool-dipoolkragte in **B**. ✓
- Compare strengths of IMFs./Vergelyk sterktes van IMKe. ✓
- Compare energies required./Vergelyk energieë benodig. ✓

- Compound A/butan-1-ol has hydrogen bonding (dipole-dipole and London forces) between molecules. ✓

- Compound B/butan-2-one has dipole-dipole forces (and London forces) between molecules. ✓

- Intermolecular forces in compound A/butan-1-ol are stronger than intermolecular forces in compound B/butan-2-one. ✓

OR

Intermolecular forces in compound B/butan-2-one are weaker than intermolecular forces in compound A/butan-1-ol. ✓

- More energy is needed to overcome/break intermolecular forces in compound A/butan-ol than in compound B/butan-2-one. ✓

- Verbinding A/butan-1-ol het waterstofbindings (dipool-dipoolkragte en Londonkragte) tussen molekule.

- Verbinding B/butan-2-oon het dipool-dipoolkragte (en London kragte) tussen molekule. ✓

- Intermolekulêre kragte in verbinding A/butan-1-ol is sterker as intermolekulêre kragte in verbinding B/butan-2-oon.

OF

Intermolekulêre kragte in verbinding B/butan-2-oon is swakker as intermolekulêre kragte in verbinding A/butan-1-ol.

- Meer energie is nodig om intermolekulêre kragte te oorkom/breek in verbinding A/butan-1-ol as in verbinding B/butan-2-oon.

(4)

3.5

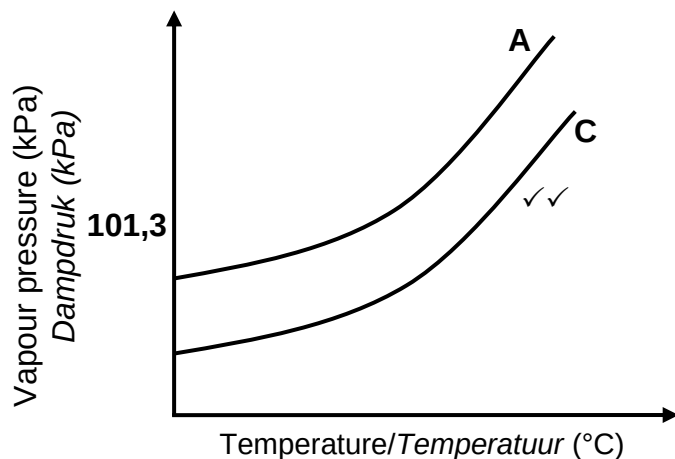
3.5.1 Boiling point (of compound **A**/butan-1-ol) ✓
Kookpunt (van verbinding **A**/butan-1-ol)

(1)

3.5.2 Gas ✓

(1)

3.5.3



Marking criteria/Nasienkriteria:

- Curve **C** starts below curve **A**/Kurwe **C** begin onder kurwe **A**. ✓
- Curve **C** remains below curve **A**/ Kurwe **C** bly onder kurwe **A**. ✓

Accept/Aanvaar

- If **C** is labelled as **B** / Indien **C** as **B** benoem is
- If graph below graph **A** is unlabelled /Indien grafiek onder grafiek **A** nie benoemis nie

Note/Let Wel

If both graphs unlabelled / Indien beide grafiek nie benoem is nie:
0 marks / 0 punte

(2)
[12]

QUESTION 4/VRAAG 4

4.1

4.1.1 Heat/sunlight/ultraviolet light/radiation/light ✓
Hitte/sonlig/ultravioletlig/straling/lig (1)

4.1.2 HBr/hydrogen bromide/waterstofbromied ✓ (1)

4.1.3 Hydrolysis/hidrolise ✓ (1)

4.1.4 H₂O/water ✓
Accept/Aanvaar
hydrogen oxide/waterstofoksied

OR/OF

NaOH/KOH/LiOH/sodium hydroxide/potassium hydroxide/lithium hydroxide
NaOH/KOH/LiOH/Natriumhidroksied/kaliumhidroksied/litiumhidroksied (1)

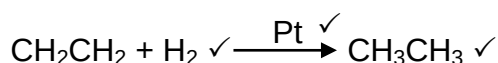
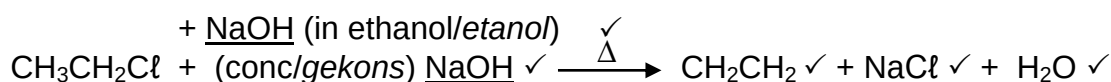
4.1.5 2-bromo ✓ propane ✓
2-bromopropaan (2)

4.2

Marking criteria/Nasienkriteria:

(Mark bullets independently. / Sien kolpunte onafhanklik na.)

- React chloroethane with (conc) NaOH or NaOH in ethanol. ✓
- Indicate heat/Δ (on the arrow) or as a reactant in the reaction of chloroethane. ✓
- Correct condensed formula for ethene as product. ✓
- Product NaCl in the reaction of chloroethane. ✓
- Product H₂O in the reaction of chloroethane. ✓
- React ethene with H₂. ✓
- Indicate Pt on the arrow of / at the reaction of ethene with H₂. ✓
- Correct condensed formula of ethane as product. ✓
- Reageer chloroetaan met (gekons) NaOH of NaOH in etanol. ✓
- Dui hitte/Δ (op die pyl) of as 'n reaktant in die reaksie van chloroetaan. ✓
- Korrekte gekondenseerde formule vir eteen as produk. ✓
- Produk NaCl in die reaksie van chloroetaan. ✓
- Produk H₂O in die reaksie van chloroetaan. ✓
- Reageer eteen met H₂. ✓
- Dui Pt aan op die pyl / by die reaksie van eteen met H₂. ✓
- Korrekte gekondenseerde formule vir etaan as produk. ✓



Note/Let wel

Any additional reactants or products: Deduct one mark per reaction
Enige addisionele reaktanse of produkte: Trek een punt af per reaksie

(8)
[14]

QUESTION 5/VRAAG 5

5.1

NOTE/LET WEL

Give the mark for per unit time only if in context of reaction rate.

Gee die punt vir per eenheidtyd slegs indien in konteks met reaksietempo.

ANY ONE/ENIGE EEN

- Change in concentration ✓ of products/reactants per (unit) time. ✓
Verandering in konsentrasie van produkte/reaktanse per (eenheid)tyd.
- Change in amount/number of moles/volume/mass of products or reactants per (unit) time.
Verandering in hoeveelheid/getal mol/volume/massa van produkte of reaktanse per (eenheid)tyd.
- Amount/number of moles/volume/mass of products formed/reactants used per (unit) time.
Hoeveelheid/getal mol/volume/massa van produkte gevorm/reaktanse gebruik per (eenheid)tyd.
- Rate of change in concentration/amount/number of moles/volume/mass.
Tempo van verandering in konsentrasie/ hoeveelheid/getal mol/ volume/ massa. ✓✓ **(2 or/lof 0)**

(2)

5.2

- Time/tyd ✓
- Volume of gas/CO₂/carbon dioxide (in gas syringe)✓
Volume gas/CO₂/koolstofdioksied (in gasspuit)

OR/OF

- Time taken for Al₂(CO₃)₃ to be used up. ✓✓
Tyd geneem vir die Al₂(CO₃)₃ om opgebruik te word.

Accept/Aanvaar

Measure volume of gas/CO₂ at regular time intervals. ✓✓

Meet volume van gas/CO₂ met gereelde tydintervalle.

(2)

5.3

Experiment II/Eksperiment II:

- More (HCl) particles per unit volume./More particles with correct orientation. ✓
- More effective collisions per unit time./Higher frequency of effective collisions. ✓
- Higher reaction rate. ✓
- Meer (HCl)-deeltjies per eenheid volume./Meer deeltjies met korrekte oriëntasie.
- Meer effektiewe botsings per eenheid tyd./Hoër frekwensie van effektiewe botsings.
- Hoër reaksietempo.

(3)

OR/OF**Experiment I/Eksperiment I:**

- Less (HCl) particles per unit volume. ✓
- Less effective collisions per unit time./Lower frequency of effective collisions. ✓
- Lower reaction rate. ✓
- Minder (HCl) deeltjies per eenheidvolume.
- Minder effektiewe botsings per eenheidtyd./ Laer frekwensie van effektiewe botsings.
- Laer reaksietempo.

5.4

<u>OPTION 1/OPSIE 1</u> ave rate/gem tempo = $-\frac{\Delta n}{\Delta t}$ $4,4 \times 10^{-3} = -\frac{n_f - 0,016}{2,5 (-0)}$ ✓ $n[\text{Al}_2(\text{CO}_3)_3] = 0,005 \text{ (mol)}$ ✓	<u>Marking criteria/Nasienkriteria</u> • Substitute average rate and Δt. / Vervang gemiddelde tempo en Δt. ✓ • Substitute/Vervang Δn. ✓ • Final answer/Finale antwoord: 0,005 (mol) ✓ <u>NOTE/LET WEL</u> • Accept negative answers when the negative sign in front of the formula is omitted. / Aanvaar negatiewe antwoord wanneer die negatiewe teken voor die formule uitgelaat is. • Do not penalise if initial and final mole values or time values are swapped. / Moenie penaliseer indien aanvanklike en finale molwaardes omgeruil is nie.
<u>OPTION 2/OPSIE 2</u> ave rate/gem tempo = $\frac{\Delta n}{\Delta t}$ $4,4 \times 10^{-3} = \frac{\Delta n}{2,5}$ ✓ $\Delta n[\text{Al}_2(\text{CO}_3)_3] = 0,016 - 0,011$ ✓ $= 0,005 \text{ mol}$ ✓	
<u>OPTION 3/OPSIE 3</u> <u>With reference to CO₂/Met verwysing na CO₂</u> ave. rate/gem tempo = $\frac{\Delta n}{\Delta t}$ $4,4 \times 10^{-3} = \frac{\Delta n}{2,5}$ ✓ $\Delta n(\text{CO}_2) = 0,011 \text{ mol}$ $n(\text{CO}_2) : n(\text{Al}_2(\text{CO}_3)_3)$ 3 : 1 0,011 : $3,67 \times 10^{-3} \text{ mol}$ ✓ $n(\text{Al}_2(\text{CO}_3)_3 \text{ left/oor}) = 0,016 - 3,67 \times 10^{-3} = 1,23 \times 10^{-2} \text{ mol}$ ✓	

OPTION 4/OPSIE 4**With reference to HCl/Met verwysing na HCl**

$$\text{ave. rate/gem tempo} = \frac{\Delta n}{\Delta t}$$

$$4,4 \times 10^{-3} = \frac{\Delta n}{2,5}$$

$$\Delta n(\text{HCl}) = 0,011 \text{ mol}$$

$$n[\text{Al}_2(\text{CO}_3)_3] = \frac{0,011}{6} = 0,0018 \text{ mol} \checkmark$$

$$n[\text{Al}_2(\text{CO}_3)_3] \text{ left/oor} = 0,016 - 0,0018 = 0,0142 \text{ mol} \checkmark$$

OPTION 5/OPSIE 5**With reference to AlCl₃/Met verwysing na AlCl₃**

$$\text{ave. rate/gem tempo} = \frac{\Delta n}{\Delta t}$$

$$4,4 \times 10^{-3} = \frac{\Delta n}{2,5}$$

$$\Delta n(\text{AlCl}_3) = 0,011 \text{ mol}$$

$$n[\text{Al}_2(\text{CO}_3)_3] = 0,0055 \text{ mol} \checkmark$$

$$n[\text{Al}_2(\text{CO}_3)_3] \text{ left/oor} = 0,016 - 0,0055 = 0,0105 \text{ mol} \checkmark$$

(3)

5.5

Marking criteria/Nasienkriteria:

- Use mol ratio/Gebruik molverhouding: $n(\text{CO}_2) : n(\text{Al}_2(\text{CO}_3)_3) = 3 : 1 \checkmark$

- Substitute $24\,000 \text{ cm}^3 \cdot \text{mol}^{-1} / 24 \text{ dm}^3 \cdot \text{mol}^{-1}$ in $n = \frac{V}{V_M}$ or in ratio. $\checkmark$

Vervang $24\,000 \text{ cm}^3 \cdot \text{mol}^{-1} / 24 \text{ dm}^3 \cdot \text{mol}^{-1}$ in $n = \frac{V}{V_M}$ of in verhouding.

- Final answer/Finale antwoord: $1\,152 \text{ cm}^3 / 1,152 \text{ dm}^3 \checkmark$

OPTION 1/OPSIE 1

$$\begin{aligned} n(\text{CO}_2) &= 3n[\text{Al}_2(\text{CO}_3)_3] \\ &= 3(0,016) \checkmark \\ &= 0,048 \text{ mol} \end{aligned}$$

$$n(\text{CO}_2) = \frac{V}{V_M}$$

$$\therefore 0,048 = \frac{V}{24000} \checkmark$$

$$V(\text{CO}_2) = 1\,152 \text{ cm}^3 (1,152 \text{ dm}^3) \checkmark$$

OPTION 2/OPSIE 2

$$\begin{aligned} n(\text{CO}_2) &= 3n[\text{Al}_2(\text{CO}_3)_3] \\ &= 3(0,016) \checkmark \\ &= 0,048 \text{ mol} \end{aligned}$$

$$1 \text{ mol} \dots\dots\dots 24\,000 \text{ cm}^3$$

$$0,048 \text{ mol} \dots\dots\dots V$$

$$V(\text{CO}_2) = \frac{0,048 \times 24000}{1} \checkmark$$

$$= 1\,152 \text{ cm}^3 (1,152 \text{ dm}^3) \checkmark$$

(3)

[13]

QUESTION 6/VRAAG 6

- 6.1 (The stage in a chemical reaction when the) rate of forward reaction equals the rate of reverse reaction. ✓✓
(Die stadium in 'n chemiese reaksie wanneer die) tempo van die voorwaartse reaksie gelyk is aan die tempo van die terugwaartse reaksie. (2 orlof 0)

OR/OF

- (The stage in a chemical reaction when the) concentrations of reactants and products remain constant.
(Die stadium in 'n chemiese reaksie wanneer die) konsentrasies van reaktanse en produkte konstant bly. (2 orlof 0) (2)

6.2

6.2.1 X ✓



ANY ONE/ENIGE EEN

- The concentration of products increases (from 0 – 6 min.).
Die konsentrasie van die produkte neem toe (van 0 - 6 min.).
- The concentration of reactants decreases (from 0 – 6 min.).
Die konsentrasie van die reaktanse neem af (van 0 – 6 min.).
- No products were present initially. ✓
Geen produkte was aanvanklik teenwoordig nie.
- The curve begins at zero./Die kurwe begin by nul. (2)

6.2.2 Higher than/Hoër as ✓ (1)

6.3 CALCULATIONS USING NUMBER OF MOLES BEREKENINGE WAT AANTAL MOL GEBRUIK

Marking criteria/Nasienkriteria

- Calculate/Bereken mol HI: $n(\text{HI})_{\text{ini/aanv.}} = 1(0,5)$. ✓
- Use mol ratio/Gebruik molverhouding: 2:1:1 / $n(\text{HI}) = 2n(\text{H}_2) = 2n(\text{I}_2)$. ✓
- $n(\text{H}_2)_{\text{equilibrium/ewewig}} = n(\text{H}_2)_{\text{formed/gevorm}}$
 $n(\text{I}_2)_{\text{equilibrium/ewewig}} = n(\text{I}_2)_{\text{formed/gevorm}}$ } ✓
- **Note:** If Δn not shown award mark for equal $n_{\text{equilibrium}}$
Let wel: Indien Δn nie aangedui is nie, ken punt toe vir gelyke n_{ewewig}
- $n((\text{HI})_{\text{equilibrium/ewewig}} = n(\text{HI})_{\text{initial/aanvanklik}} - n(\text{HI})_{\text{change/verandering}}$. ✓
- Divide $n(\text{HI})_{\text{equil}}$ & $n(\text{H}_2)_{\text{equil}}$ & $n(\text{H}_2)_{\text{equil}}$ by $0,5 \text{ dm}^3$. ✓
Deel $n(\text{HI})_{\text{ewewig}}$ & $n(\text{H}_2)_{\text{ewewig}}$ & $n(\text{H}_2)_{\text{ewewig}}$ deur $0,5 \text{ dm}^3$.
- Correct K_c expression (formulae in square brackets). ✓
Korrekte K_c -uitdrukking (formules in vierkanthakies).
- Substitute 0,04 into K_c expression. ✓
Vervang 0,04 in K_c -uitdrukking.
- Substitute equilibrium concentrations in K_c expression. ✓
Vervang ewewigskonsentrasies in K_c -uitdrukking.
- Final answer/Finale antwoord: 0,07 mol ✓
Range/Gebied: 0,07 – 0,072 mol

OPTION 1/OPSIE 1

$$n(\text{HI}) = 1(0,5) = 0,5 \text{ mol}$$

	HI	H ₂	I ₂
Initial quantity (mol) <i>Aanvangshoeveelheid (mol)</i>	0,5 ✓	0	0
Change (mol) <i>Verandering (mol)</i>	2x	x	x
Quantity at equilibrium (mol)/ <i>Hoeveelheid by ewewig (mol)</i>	0,5-2x ✓	x	x ✓
Equilibrium concentration (mol·dm ⁻³) <i>Ewewigskonsentrasie (mol·dm⁻³)</i>	$\frac{0,5-2x}{0,5}$	$\frac{x}{0,5}$	$\frac{x}{0,5}$

ratio ✓
verhoudingdivide by 0,5 ✓
deel deur 0,5

$$K_c = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2} \checkmark$$

$$0,04 = \frac{\left(\frac{x}{0,5}\right)\left(\frac{x}{0,5}\right)}{\left(\frac{0,5-2x}{0,5}\right)^2} \checkmark$$

$$x = 0,071 \text{ mol} \checkmark$$

No K_c expression, correct substitution/Geen K_c-
uitdrukking, korrekte substitusie: Max./Maks. $\frac{8}{9}$

Wrong K_c expression /Verkeerde K_c-uitdrukking:
Max./Maks. $\frac{6}{9}$

CALCULATIONS USING CONCENTRATION**BEREKENINGE WAT KONSENTRASIE GEBRUIK****Marking criteria/Nasienkriteria:**

- Use initial/Gebruik aanvanklike $c(\text{HI}) = 1 \text{ mol} \cdot \text{dm}^{-3}$. ✓
- Use mol ratio/Gebruik molverhouding: 2 : 1 : 1 / $n(\text{HI}) = 2n(\text{H}_2) = 2n(\text{I}_2)$. ✓
- $c(\text{H}_2)_{\text{equilibrium/ewewig}} = c(\text{H}_2)_{\text{formed/gevorm}}$ } ✓
 $c(\text{I}_2)_{\text{equilibrium/ewewig}} = c(\text{I}_2)_{\text{formed/gevorm}}$
- **Note:** If Δc not shown award mark for equal $c_{\text{equilibrium}}$
Let wel: Indien Δc nie aangedui is nie, ken punt toe vir gelyke c_{ewewig}
- $c(\text{HI})_{\text{equilibrium/ewewig}} = c(\text{HI})_{\text{initial}} - c(\text{HI})_{\text{change}}$. ✓
- Correct K_c expression (formulae in square brackets). ✓
 Korrekte K_c-uitdrukking (formules in vierkanthakies).
- Substitution of 0,04 into K_c expression. ✓
 Vervang 0,04 in K_c-uitdrukking.
- Substitution of equilibrium concentrations into K_c expression. ✓
 Vervanging van ewewigskonsentrasies in K_c-uitdrukking.
- Multiply concentration by 0,5 dm³. ✓
 Vermenigvuldig konsentrasie met 0,5 dm³.
- Final answer/Finale antwoord: 0,07 mol ✓
 Range/Gebied: 0,07 to/tot 0,072 mol

OPTION 2/OPSIE 2

	HI	H ₂	I ₂
Initial concentration (mol·dm ⁻³) <i>Aanvangskonsentrasie (mol·dm⁻³)</i>	1 ✓	0	0
Change (mol·dm ⁻³) <i>Verandering (mol·dm⁻³)</i>	2x	x	x
Equilibrium concentration (mol·dm ⁻³) <i>Ewewigskonsentrasie (mol·dm⁻³)</i>	1-2x ✓	x	x ✓

$$K_c = \frac{[H_2][I_2]}{[HI]^2} \quad \checkmark$$

$$0,04 = \frac{(x)(x)}{(1-2x)^2} \quad \checkmark$$

$$x = 0,143 \text{ mol·dm}^{-3}$$

$$n(I_2) = cV$$

$$= 0,143 \times 0,5 \quad \checkmark$$

$$= 0,072 \text{ mol} \quad \checkmark$$

No K_c expression, correct substitution/Geen K_c-uitdrukking, korrekte substitusie: Max./Maks. $\frac{8}{9}$

Wrong K_c expression /Verkeerde K_c-uitdrukking:
Max./Maks. $\frac{6}{9}$

(9)

6.4

6.4.1 Both forward and reverse/Beide voorwaartse en terugwaartse ✓

(1)

6.4.2 Positive/Positief ✓

- The forward reaction is favoured. ✓
Die voorwaartse reaksie word bevoordeel.
- An increase in temperature favours the endothermic reaction. ✓
'n Toename in temperatuur bevoordeel die endotermiese reaksie.
- The forward reaction is endothermic. ✓
Die voorwaartse reaksie is endotermies.

(4)

[19]

QUESTION 7/VRAAG 7

7.1 Standard solution/Standaardoplossing ✓

(1)

7.2

7.2.1

Marking criteria/Nasienkriteria:

- Any one of the formulae/*Enige een van die formules*: $c = \frac{m}{MV}$ / $n = \frac{m}{M}$ / $c = \frac{n}{V}$ ✓
- Substitution of $40 \text{ g} \cdot \text{mol}^{-1}$ into correct formula. ✓
Vervanging van $40 \text{ g} \cdot \text{mol}^{-1}$ in korrekte formule.
- Substitution of $0,25 \text{ dm}^3$ into correct formula. ✓
Vervanging van $0,25 \text{ dm}^3$ in korrekte formule.
- Final answer/*Finale antwoord*: $0,2 \text{ mol} \cdot \text{dm}^{-3}$ ✓

OPTION 1/OPSIE 1

$$c = \frac{m}{MV} \checkmark$$

$$= \frac{2}{\sqrt{40 \times 0,25}} \checkmark$$

$$= 0,20 \text{ mol} \cdot \text{dm}^{-3} \checkmark$$

OPTION 2/OPSIE 2

$$n = \frac{m}{M}$$

$$= \frac{2}{40} \checkmark$$

$$= 0,05 \text{ mol}$$

$$c = \frac{n}{V}$$

$$= \frac{0,05}{0,25} \checkmark$$

$$= 0,20 \text{ mol} \cdot \text{dm}^{-3} \checkmark$$

Any one formula/*enige formule* ✓

(4)

7.2.2 POSITIVE MARKING FROM 7.2.1./POSITIEWE NASIEN VAN 7.2.1.

OPTION 1/OPSIE 1

$$[\text{H}_3\text{O}^+][\text{OH}^-] = 1 \times 10^{-14}$$

$$\frac{[\text{H}_3\text{O}^+](0,2)}{[\text{H}_3\text{O}^+]} = 1 \times 10^{-14} \checkmark$$

$$[\text{H}_3\text{O}^+] = 5 \times 10^{-14} \text{ mol} \cdot \text{dm}^{-3}$$

$$\downarrow$$

$$\text{pH} = -\log[\text{H}_3\text{O}^+] \checkmark$$

$$= -\log(5 \times 10^{-14}) \checkmark$$

$$= 13,30 \checkmark$$

OPTION 2/OPSIE 2

$$\text{pOH} = -\log[\text{OH}^-] \checkmark$$

$$= -\log(0,2) \checkmark$$

$$= 0,6989 \quad (0,7)$$

$$\downarrow$$

$$\text{pH} + \text{pOH} = 14$$

$$\text{pH} = 14 - 0,6989 \checkmark$$

$$= 13,30 \checkmark$$

(4)

7.3 **POSITIVE MARKING FROM QUESTION 7.2.**
POSITIEWE NASIEN VANAF VRAAG 7.2.

Marking criteria/Nasienkriteria:

- Substitution to calculate $n(\text{NaOH})$. / Vervanging om $n(\text{NaOH})$ te bereken. ✓
- Use mol ratio / Gebruik molverhouding: $n(\text{HCl})_{\text{excess/oormaat}} : n(\text{NaOH}) = 1 : 1$. ✓
- Substitute / Vervang $100 \text{ g} \cdot \text{mol}^{-1}$ in $n = \frac{m}{M}$ ✓
- Use mol ratio / Gebruik molverhouding: $n(\text{HCl})_{\text{reacted/oormaat}} : n(\text{CaCO}_3) = 2 : 1$. ✓
- $n(\text{HCl})_{\text{initial/aanvanklik}} = n(\text{HCl})_{\text{excess/oormaat}} + n(\text{HCl})_{\text{reacted/reageer}}$ ✓✓
- Substitute $0,05 \text{ dm}^3$ to calculate either $c(\text{HCl})_{\text{initial}}$ or $c(\text{HCl})_{\text{reacted}}$ ✓
Vervang $0,05 \text{ dm}^3$ om $c(\text{HCl})_{\text{aanvanklik}}$ of $c(\text{HCl})_{\text{reageer}}$ te bereken.
- Final answer / Finale antwoord: $0,7 \text{ mol} \cdot \text{dm}^{-3}$ ✓
Range / Gebied: $0,70$ to / tot $0,90 \text{ mol} \cdot \text{dm}^{-3}$

OPTION 1 / OPSIE 1

$$\begin{aligned} n(\text{NaOH})_{\text{used/gebruik}} &= c_b V_b \\ &= 0,2 \times 0,025 \checkmark \\ &= 5 \times 10^{-3} \text{ mol} \end{aligned}$$

OPTION 2 / OPSIE 2

$$\begin{aligned} n(\text{NaOH})_{\text{used/gebruik}} &= \frac{25}{250} \times \frac{2}{40} \checkmark \\ &= 5 \times 10^{-3} \text{ mol} \end{aligned}$$

$$n(\text{HCl})_{\text{excess/oormaat}} = n(\text{NaOH}) = 5 \times 10^{-3} \text{ mol} \checkmark$$

$$\begin{aligned} n(\text{CaCO}_3) &= \frac{m}{M} \\ &= \frac{1,5}{100} \checkmark \\ &= 0,015 \text{ mol} \quad (0,02 \text{ mol}) \end{aligned}$$

$$n(\text{HCl})_{\text{reacted/reageer}} = 2n(\text{CaCO}_3) = 0,03 \text{ mol} \checkmark \quad (0,04 \text{ mol})$$

$$\begin{aligned} n(\text{HCl})_{\text{ini/aanv.}} &= 5 \times 10^{-3} + 0,03 \checkmark \checkmark \\ &= 0,035 \text{ mol} \quad (0,045 \text{ mol}) \end{aligned}$$

$$\begin{aligned} c(\text{HCl})_{\text{ini/aanv.}} &= \frac{n}{V} \\ &= \frac{0,035}{0,05} \checkmark \\ &= 0,70 \text{ mol} \cdot \text{dm}^{-3} \checkmark \quad (0,90 \text{ mol} \cdot \text{dm}^{-3}) \end{aligned}$$

OPTION 3/OPSIE 3	OPTION 4/OPSIE 4
$\frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}$ $\frac{c_a (0,05)}{(0,2)(0,025)} = \frac{1}{1} \checkmark$ $c_a = c(\text{HCl})_{\text{excess/oormaat}}$ $= 0,1 \text{ mol} \cdot \text{dm}^{-3}$ $n(\text{CaCO}_3) = \frac{m}{M}$ $= \frac{1,5}{100} \checkmark$ $= 0,015 \text{ mol}$ $n(\text{CaCO}_3) : n(\text{HCl}) = 1 : 2$ $n(\text{HCl})_{\text{reacted/reageer}} = 2(0,015) \checkmark$ $= 0,03 \text{ mol}$ $c(\text{HCl})_{\text{reacted/reageer}} = \frac{n}{V}$ $= \frac{0,03}{0,05} \checkmark$ $= 0,6 \text{ mol} \cdot \text{dm}^{-3}$ $c(\text{HCl})_{\text{initial/aanvanklik}} = c(\text{HCl})_{\text{reacted/reageer}} + c(\text{HCl})_{\text{excess/oormaat}}$ $= 0,6 + 0,1 \checkmark \checkmark$ $= 0,7 \text{ mol} \cdot \text{dm}^{-3} \checkmark$	$(\text{NaOH})_{\text{used/gebruik}} = c_b V_b$ $= (0,2)(0,025) \checkmark$ $= 0,005 \text{ mol}$ $n(\text{HCl})_{\text{excess/oormaat}} = n(\text{NaOH}) \checkmark$ $= 0,005 \text{ mol}$ $c(\text{HCl})_{\text{excess/oormaat}} = \frac{0,005}{0,05}$ $= 0,1 \text{ mol} \cdot \text{dm}^{-3}$

(8)
[17]

QUESTION 8/VRAAG 8

8.1

8.1.1 Gain of electrons./Opneem van elektrone. ✓✓ (2 or/of 0) (2)

8.1.2 $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ ✓✓

Ignore phases./Ignoreer fases.

Marking criteria/Nasienkriteria:

- $\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \leftarrow 2\text{H}_2\text{O}(\text{l}) + 2\text{e}^-$ ($\frac{2}{2}$)
 $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ ($\frac{1}{2}$)
 $\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightleftharpoons 2\text{H}_2\text{O}(\text{l}) + 2\text{e}^-$ ($\frac{0}{2}$)
 $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \leftarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ ($\frac{0}{2}$)
- Ignore if charge omitted on electron./Ignoreer indien lading weggelaat op elektron.
- If charge (-) omitted on OH^- /Indien lading (-) weggelaat op OH^- :
 Example/Voorbeeld: $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}(\text{aq})$ ✓ Max./Maks: $\frac{1}{2}$

(2)

8.1.3 $2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \checkmark \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) + 2\text{Na}^+(\text{aq}) \checkmark$ Bal ✓

OR/OF

$2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \checkmark \rightarrow \text{H}_2(\text{g}) + 2\text{NaOH}(\text{aq}) \checkmark$ Bal ✓

Ignore phases./Ignoreer fases.

Marking criteria/Nasienkriteria:

- Reactants ✓ Products ✓ Balancing ✓
 Reaktanse Produkte Balansering
- Ignore double arrows./Ignoreer dubbelpyle.
- Ignore phases./Ignoreer fases.
- Marking rule 6.3.10./Nasienreël 6.3.10.

(3)

8.1.4 Formation of hydroxide ions / OH^- / sodium hydroxide/base/ alkaline/ $\text{pH} > 7$ ✓
 Vorming van hidroksied / OH^- / natriumhidroksied / basis / alkalies / $\text{pH} > 7$ (1)

8.1.5 Cu is a weaker reducing agent ✓ than H_2 (and OH^-) ✓ and H_2O will not be reduced ✓ (to H_2 and OH^-).
 Cu is 'n swakker reduseermiddel as H_2 (and OH^-) en H_2O sal nie gereduseer word nie na H_2 (en OH^-).

OR/OF

H_2 (and OH^-) are stronger reducing agent ✓ than Cu and H_2O ✓ will not be reduced ✓ (to H_2 and OH^-).
 H_2 (en OH^-) is 'n sterker reduseermiddel as Cu en H_2O sal nie gereduseer word (na H_2 en OH^-).

(3)

8.2

8.2.1 Phase separator/boundary/difference ✓
Fase skeiding/grens/verskil

(1)

8.2.2 Chemical (energy) to electrical (energy) ✓
Chemiese (energie) na elektriese (energie)

(1)

8.2.3

OPTION/OPSIE 1 $E_{\text{cell}}^{\theta} = E_{\text{reduction}}^{\theta} - E_{\text{oxidation}}^{\theta} \checkmark$ $= 0,77 \checkmark - (-0,13) \checkmark$ $E_{\text{cell}}^{\theta} = 0,90 \text{ V} \checkmark$	Notes/Aantekeninge - Accept any other correct formula from the data sheet./Aanvaar enige ander korrekte formule vanaf gegewensblad. - Any other formula using unconventional abbreviations, e.g. $E_{\text{cell}}^{\theta} = E_{\text{OA}}^{\theta} - E_{\text{RA}}^{\theta}$ followed by correct substitutions:/Enige ander formule wat onkonvensionele afkortings gebruik bv. $E_{\text{sel}}^{\theta} = E_{\text{OM}}^{\theta} - E_{\text{RM}}^{\theta}$ gevolg deur korrekte vervangings: Max/Maks: $\frac{3}{4}$
OPTION/OPSIE 2 $\begin{array}{lcl} \checkmark \left\{ \begin{array}{l} \text{Pb(s)} \rightarrow \text{Pb}^{2+}(\text{aq}) + 2\text{e}^{-} \\ 2\text{Fe}^{3+}(\text{aq}) + 2\text{e}^{-} \rightarrow 2\text{Fe}^{2+}(\text{aq}) \end{array} \right. & & \begin{array}{l} 0,13 \text{ (V)} \checkmark \\ 0,77 \text{ (V)} \checkmark \end{array} \\ \hline \text{Pb}^{2+}(\text{aq}) + 2\text{Fe}^{3+}(\text{aq}) \rightarrow \text{Pb(s)} + 2\text{Fe}^{2+}(\text{aq}) & & 0,90 \text{ V} \checkmark \end{array}$	

(4)

[17]

QUESTION 9/VRAAG 9

9.1 Electrolytic (cell)/Elektrolitiese (sel) ✓



Cells have a battery/DC power source/ Electrical energy is converted to chemical energy. ✓

Selle het batterye/GS kragbron/ Elektriese energie is omgeskakel na chemiese energie.

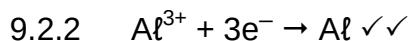
(2)

9.2

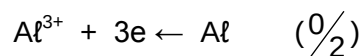
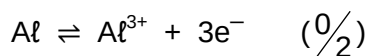
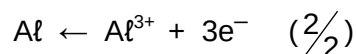
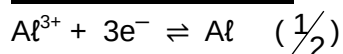
9.2.1 $2\text{Cl}^{-} \rightarrow \text{Cl}_2 + 2\text{e}^{-} \checkmark \checkmark$

Notes/Aantekeninge $2\text{Cl}^{-} \rightleftharpoons \text{Cl}_2 + 2\text{e}^{-} \left(\frac{1}{2} \right) \qquad \text{Cl}_2 + 2\text{e}^{-} \leftarrow 2\text{Cl}^{-} \left(\frac{2}{2} \right)$ $\text{Cl}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{Cl}^{-} \left(\frac{0}{2} \right) \qquad 2\text{Cl}^{-} \leftarrow \text{Cl}_2 + 2\text{e}^{-} \left(\frac{0}{2} \right)$ - Ignore if charge omitted on electron./Ignoreer indien lading weggelaat op elektron. - If charge (-) omitted on Cl^{-}/Indien lading (-) weggelaat op Cl^{-}: Example/Voorbeeld: $2\text{Cl}(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^{-}$ Max./Maks: $\frac{1}{2}$	
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(2)



Notes/Aantekeninge



- Ignore if charge omitted on electron./Ignoreer indien lading weggelaat op elektron.
- If charge (+) omitted on Al^{3+} /Indien lading (+) weggelaat op Al^{3+} :

Example/Voorbeeld: $\text{Al}^3(\text{aq}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$ Max./Maks: $\frac{1}{2}$

(2)



(1)

9.3 **ANY ONE/ENIGE EEN**

- The electrode/carbon/C reacts with oxygen. ✓
Die elektrode/koolstof/C reageer met suurstof.
- $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- Oxidation takes place./Electrons are lost.
Oksidasie vind plaas./Elektrone word verloor.
- Oxygen corrodes the carbon electrode.
Suurstof roes die koolstof elektrode.

(1)

[8]

QUESTION 10/VRAAG 10

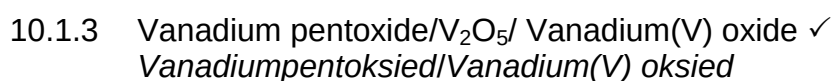
10.1



(1)



(1)



(1)



Marking guidelines/Nasienkriteria:

- Reactants ✓ Products ✓ Balancing ✓
Reaktanse ✓ Produkte ✓ Balansering ✓
- Ignore/Ignoreer → and phases / en fases
- Marking rule 6.3.10/Nasienreël 6.3.10

(3)

10.2

10.2.1 The ratio of nitrogen (N), phosphorous (P) and potassium (K) in a fertiliser./The ratio of the primary nutrients ✓
Die verhouding van stikstof (N), fosfor (P) en kalium (K) in die kunsmis. / Die verhouding van primêre nutriënte.

(1)

10.2.2

OPTION 1/OPSIE 1

Mass N in 4 kg NH_4NO_3 / Massa N in 4 kg NH_4NO_3

$$m(\text{N}) = \frac{28}{80} \times 4 \checkmark$$

$$= 1,4 \text{ kg}$$

$$m(\text{K}) = 2m(\text{N}) \checkmark$$

$$= 2,8 \text{ kg}$$

$$m(\text{P}) = 3m(\text{N}) \checkmark$$

$$= 4,2 \text{ kg}$$

$$m(\text{fertiliser/kunsmis}) = 1,4 + 2,8 + 4,2 \\ = 8,4 \text{ kg} \checkmark$$

OPTION 2/OPSIE 2

Mass N in 4 kg NH_4NO_3 / Massa N in 4 kg NH_4NO_3 :

$$m(\text{N}) = \frac{28}{80} \times 4 \checkmark$$

$$= 1,4 \text{ kg}$$

N : P : K

1 : 3 : 2

$$\therefore m(\text{fertiliser/kunsmis}) = (6) \checkmark (1,4) \checkmark \\ = 8,4 \text{ kg} \checkmark$$

OPTION 3/OPSIE 3

$$\% \text{ N} = \frac{(2)(14)}{80} \times 100 = 35\%$$

$$\text{Nitrogen in 4 kg} = 35\% \text{ of/van } 4 = 1,4 \text{ kg} \checkmark$$

N : P : K

1 : 3 : 2

1,4 : 4,2 $\checkmark$: 2,8 $\checkmark$

$$\text{Total mass of fertiliser / Totale massa kunsmis} = 1,4 + 4,2 + 2,8 \\ = 8,4 \text{ kg} \checkmark$$

(4)
[11]

TOTAL/TOTAAL: 150